Ali Mehdi Khan

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Professional Summary

Computer Science graduate (B.Tech, 2026) and Junior Software Developer at imaPRO building speech, NLP and LLM applications with Python and FastAPI. Built and deployed a full-stack AI pronunciation coach with scoring validated against expert ratings ($r = 0.70$; 0.94 AUC for mispronunciation detection). Skilled in machine learning, model evaluation, ONNX optimization, Docker, CI/CD and Google Cloud (Vertex AI, Gemini).

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, SQL, C, C++
AI / ML: Machine Learning, Deep Learning, PyTorch, TensorFlow, Keras, Hugging Face Transformers, ONNX Runtime, faster-whisper, wav2vec 2.0, NLP, Speech Recognition (ASR), Computer Vision (CNNs), Model Evaluation
LLMs: Prompt Engineering, LLM Evaluation, Groq API, OpenAI API, Gemini (Vertex AI); familiar with LangChain, LangGraph, RAG Pipelines, FAISS, Pinecone
Web: FastAPI, REST APIs, SQLAlchemy, PostgreSQL, SQLite, React, SvelteKit, Next.js, Tailwind CSS
DevOps: Docker, GitHub Actions (CI/CD), pytest, Hugging Face Spaces, Google Cloud, Prometheus, Git, Linux

Experience

Junior Software Developer (Web & Mobile), imaPRO, Lucknow — Aug 2026 – Present

- Leading the migration of a production web app's frontend to SvelteKit 5, rebuilding pages as reusable reactive components; kept the backend API contract unchanged and verified each migrated route, with no regressions in booking and data flows.
- Built and tested the booking flow end to end, from the UI through the Hono REST API to the SQLite database.
- Implemented responsive layouts and shared UI components for consistent behavior on mobile and desktop.

Machine Learning Intern, GrasTech, Lucknow — Jun 2025 – Jul 2025

- Built skin cancer detection and diabetes prediction models in Python (TensorFlow/Keras), applying deep learning, feature engineering and hyperparameter tuning to real-world medical data.

Projects

AI Pronunciation Coach | *Python, FastAPI, faster-whisper, wav2vec 2.0, ONNX Runtime, React, PostgreSQL, Docker*
Live Demo: huggingface.co/spaces/Alimehdi973/ai-pronunciation-coach
Code: github.com/alimehdikhan/A.I-Pronunciation-Coach

- Built a full-stack pronunciation coach: faster-whisper transcribes speech and a wav2vec 2.0 phoneme model scores every sound using alignment-free goodness-of-pronunciation (GOP) scoring from an Interspeech 2024 paper.
- Validated against expert ratings on 500 held-out speechocean762 recordings, beating the Whisper-only baseline: sentence-score correlation 0.51 to 0.70, fluency -0.08 to 0.63, and 0.94 AUC for mispronunciation detection.
- Exported the phoneme model from PyTorch to int8 ONNX and published it on Hugging Face Hub; it scores a 4-second clip in about 1 second on a 4-core CPU.
- Added an LLM coaching layer (Groq, gpt-oss-120b) that explains only scorer-diagnosed errors, with output validation, caching and a rule-based fallback; benchmarked reasoning-effort settings (medium: about 2 s vs 18 s, no quality gain).

Skin Cancer Detection System | *Python, TensorFlow, Keras, CNN, Deep Learning, Computer Vision*
Code: github.com/alimehdikhan/Cancer-Detection-Model

- Trained a binary CNN classifier on skin lesion images for early skin cancer detection, reaching 90%+ accuracy; reduced overfitting with data augmentation and a systematic hyperparameter search.

Education

Bachelor of Technology (B.Tech), Computer Science and Engineering — 2022 – 2026
Babu Banarasi Das University (BBDU), Lucknow

Certifications

Google Cloud Skill Badges: Build Real-World AI Apps with Gemini & Imagen; Prompt Design in Vertex AI (2025)
freeCodeCamp: Machine Learning with Python (2025) | **Deloitte (Forage):** Technology Job Simulation (2025)

Volunteering

Programming Instructor (Volunteer), Tech for Good, Lucknow — 2025
Taught programming to 30+ underprivileged students over 3 months; each one built a first working application.